Chapter 12:

- Terminals:
  ↳ Letters in an alphabet

- Non terminals:
  ↳ Placeholders

Non-terminals

$S \rightarrow \{S\}$
$S \rightarrow \Lambda$

Terminals

Production

$\Sigma = \{\{, \}\}$

Example,
$S \rightarrow 0S1 \mid SS \mid \Lambda$

0 0 1 0 0 1 1 0 1 1 0 0 1 1

find nearest match

- Generate {}

$S \Rightarrow \{S\}$
$\Rightarrow \{\Lambda\}$
$\Rightarrow \{\}$

Left Derivation

Q1. $S \rightarrow aS \mid bb$ $\quad a^*bb$

↳ Replace S by infinite aS equaling to a^* & then choose bb to stop.

Q2.

$S \rightarrow XYX$

$X \rightarrow aX \mid bX \mid \Lambda$

$Y \rightarrow bbb$

aX & bX allows to generate any combination of a, b & then Λ helps stop OR just give Null string equaling to $(a+b)^*$

Y gives triple b

Q3.

(i) $S \to aX$

$X \to aX \mid bX \mid \Lambda$

$a(a+b)^*$

(ii) $S \to XaXaX$

$X \to aX \mid bX \mid \Lambda$

$(a+b)^* a (a+b)^* a (a+b)^*$

Q4.

$S \to SS \mid XaXaX \mid \Lambda$

$X \to bX \mid \Lambda$

(i) Λ, b, bb... can be generated by repeatedly replace X by bX to get b^*

(ii) Just put in S, $b^*ab^*ab^*$

(iii) Inclusion of SS allows to repeatedly get $b^*ab^*ab^*$

(iv) We don't have b^* else we have even as

(v) This allows b^*

Q5.

$S \to XbaaX \mid aX$

$X \to Xa \mid Xb \mid \Lambda$

(i) $(a+b)^* baa (a+b)^* + a(a+b)^*$

(ii) Have substring baa or begin a or both

$S \to XbaaX$

$\Rightarrow \Lambda baaX$

$S \to Xa$

$\Rightarrow Xaa$

$\Rightarrow Xbaa$

$\Rightarrow$ baaΛ

$\Rightarrow$ baa

$\Rightarrow$ Λbaa

$\Rightarrow$ baa

Q7.

(a)

$S \to aX$

$X \to bX \mid \Lambda$

(b)

$S \to XY$

$X \to aX \mid \Lambda$

$Y \to bY \mid \Lambda$

(c)

$S \to baaS \mid abbS \mid \Lambda$

Q10.

Words where bs in multiples of 3

Q11. $S \to EAE \mid SS$

$E \to EaEbE \mid EbEaE \mid \Lambda$

$A \to a$

Equal as & bs

$S \to SaSbS \mid SbSaS \mid \Lambda$

. . . . a . . . . b . . . .

. . . . b . . . . a . . . .

Each seq has a# = b# so this is recursive calls

Q15.

(i)

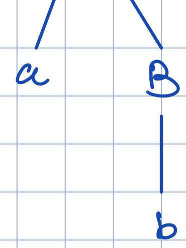
CFG1: S

CFG2: NO

CFG3: NO

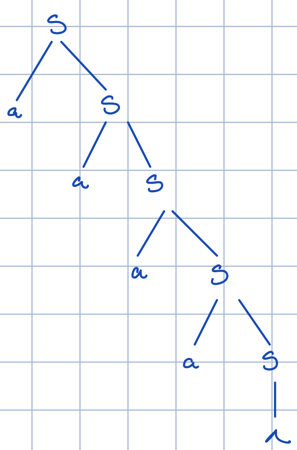
CFG4: NO

CF5: S



(ii)

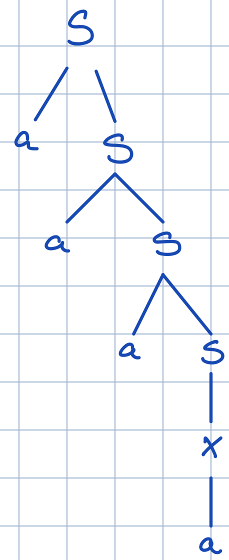
CFG2:



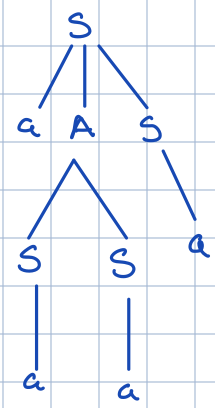
CFG1:

No

CFG3:



CFG4:

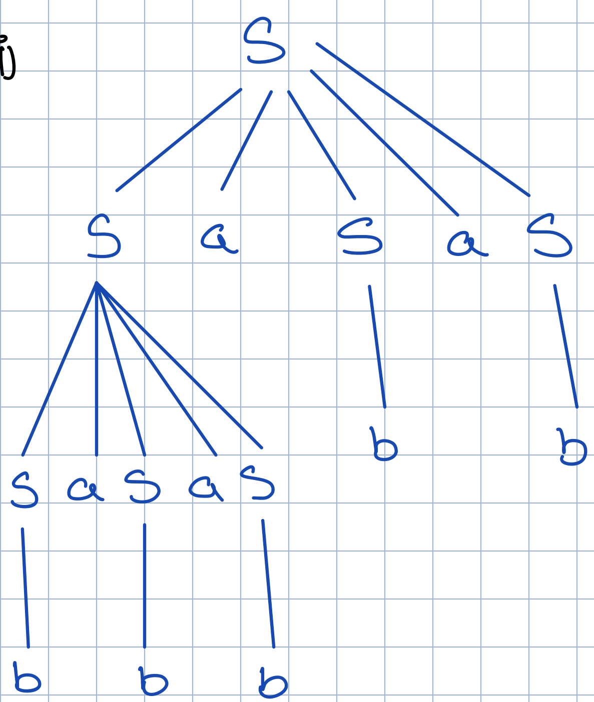


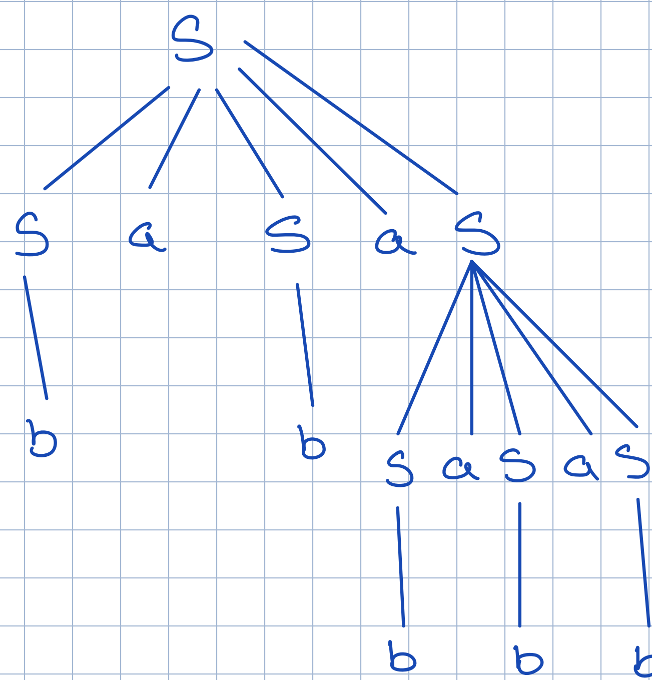
CFG5:

No

Q16.

(i)





Q8.

(i)

$S \to aS \mid bX \mid \Lambda$

$X \to aS \mid bY \mid \Lambda$

$Y \to aS \mid \Lambda$

(ii)

$S \to aS \mid bB1$

$B1 \to aB1 \mid bB2$

$B2 \to aB2 \mid bB3 \mid \Lambda$

(iii)

$S \to aA \mid bB \mid \Lambda$

$A \to aA \mid \Lambda$

$B \to bB \mid \Lambda \mid aA$

$B3 \to aB3 \mid \Lambda$

(iv)

$S \to aS \mid bA \mid \Lambda$

$A \to bA \mid aB \mid \Lambda$

$B \to bA \mid \Lambda$

(v)

$S \to aAb \mid bAa$

$A \to aA \mid bA \mid \Lambda$

Chapter 13

(i) $S \to aaaS \mid bS \mid \Lambda$

(ii) $S \to ABA$

$A \to aA \mid bA \mid \Lambda$

$B \to bbb \mid aaa$

(iii) $S \to bB \mid aA \mid \Lambda$

$A \to aB \mid bS \mid \Lambda$

$B \to bS \mid \Lambda$

(iv) $S \to aS \mid bO$

$O \to aO \mid bE$

$E \to aS \mid bO \mid \Lambda$

(v) $S \to aO \mid bO$

$O \to \Lambda \mid aE \mid bE$

$E \to aO \mid bO$

(vi) $S \to A \mid B$

$A \to bA \mid Ab \mid a$

$B \to aA \mid Aa \mid b$

(vii) $S \to X \mid Y$

$X \to aO \mid bX$

$Y \to bE \mid aY \mid \Lambda$

$O \to aX \mid \Lambda \mid bO$

$E \to bY \mid aE$

Q3.

(i) $S \to aX \mid bS \mid a \mid b$

$X \to aX \mid a$

$b^*(a+b)a^*$

(ii) $S \to bS \mid aX \mid b$

$X \to bX \mid aS \mid a$

Balanced Paranthesis: $S \to (S) \mid SS \mid \Lambda$

Palindrome: $S \to aSa \mid bSb \mid a \mid b \mid \Lambda$

Even Palindrome: $S \to aSa \mid bSb \mid \Lambda$

Odd Palindrome: $S \to aSa \mid bSb \mid a \mid b$

Equal as & bs: $S \to SaSbS \mid SbSaS \mid \Lambda$

$a^n b^n$: $S \to aSb \mid \Lambda$

$a^n b^m c^n$: $S \to aSc \mid B$

$B \to bB \mid \Lambda$

$a^n b^n c^m$: $S \to aAbX$

$A \to aAb \mid \Lambda$

$X \to cX \mid \Lambda$

Start, middle, end same letter: $S \to X \mid a \mid b$

$X \to aM_A a \mid bM_B b$

$M_A \to CM_A C \mid a$

$C \to a \mid b$

$M_B \to cM_B c \mid b$